ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 75	$4x + 60 = 360$
2	**(c)** 20 cm	$2k(3 + 5) = 64$ gives $k = 4$
3	**(a)** 34 cm	AO = OC; find x first, then AC = 2 AO
4	**(b)** 17	Adjacent angles are supplementary
5	**(b)** 85°	$\angle B = \angle D$ and the angle sum is 360°
6	**(b)** 85°	Exterior angles of a quadrilateral add up to 360°
7	**(a)** rectangle	A parallelogram with equal diagonals
8	**(d)** 60	$4x + 120 = 360$
9	**(b)** 18 cm	$2k(2 + 3) = 60$ gives $k = 6$
10	**(c)** 20 cm	AO = OC; find x first, then AC = 2 AO

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One part $= 360 \div 18 = 20^\circ$ (1); largest $= 160^\circ$ (1).	1+1
12	AO = OC gives $x = 4$ (1); AO $= 17$ cm so AC $= 34$ cm (1).	1+1
13	Fourth exterior $= 360 - 270 = 90^\circ$ (1); interior $= 180 - 90 = 90^\circ$ (1).	1+1
14	One part $= 360 \div 18 = 20^\circ$ (1); largest $= 120^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4x + 90 = 360$ (1); $x = 67$ (1); angles 67°, 82°, 97°, 112° (1).	1+1+1
16	Sides $= 2k$ and $5k$; $2(2k + 5k) = 56$ (1); $k = 4$ (1); sides 8 cm and 20 cm (1).	1+1+1
17	$\angle CAB = 90 - 25 = 65^\circ$ (1); OA = OB so $\angle AOB = 180 - 2(65) = 50^\circ$ (1); OC = OD so $\angle ODC = \angle OCD = \angle CAB = 65^\circ$ (1).	1+1+1
	OR: $4x + 48 = 360$ (1); $x = 78$ (1); angles 78°, 86°, 94°, 102° (1).	1+1+1
18	Sides $= 3k$ and $4k$; $2(3k + 4k) = 84$ (1); $k = 6$ (1); sides 18 cm and 24 cm (1).	1+1+1
	OR: $\angle CAB = 90 - 34 = 56^\circ$ (1); OA = OB so $\angle AOB = 180 - 2(56) = 68^\circ$ (1); OC = OD so $\angle ODC = \angle OCD = \angle CAB = 56^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diagonals meet at right angles, so $\angle OAB + \angle OBA = 90^\circ$ (1); $2x + 10 + 3x - 5 = 90$ gives $x = 17$ (2); $\angle A = 2 \times 44 = 88^\circ$ (1); $\angle B = 2 \times 46 = 92^\circ$ (1).	1+2+1+1
	OR: Diagonals of a rectangle are equal: $3x + 5 = 2x + 15$ gives $x = 10$ (2); AC = BD $= 35$ cm (1); OA $= 17.5$ cm (1); wire $= 2 \times 35 = 70$ cm (1).	2+1+1+1
20	Diagonals meet at right angles, so $\angle OAB + \angle OBA = 90^\circ$ (1); $3x + 5 + 2x - 5 = 90$ gives $x = 18$ (2); $\angle A = 2 \times 59 = 118^\circ$ (1); $\angle B = 2 \times 31 = 62^\circ$ (1).	1+2+1+1
	OR: Diagonals of a rectangle are equal: $5x + 2 = 3x + 12$ gives $x = 5$ (2); AC = BD $= 27$ cm (1); OA $= 13.5$ cm (1); wire $= 2 \times 27 = 54$ cm (1).	2+1+1+1

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).

- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 14	Adjacent angles are supplementary
2	**(b)** 95°	$\angle B = \angle D$ and the angle sum is 360°
3	**(b)** 95°	Exterior angles of a quadrilateral add up to 360°
4	**(d)** kite	Only one diagonal is bisected
5	**(b)** 45	$4x + 180 = 360$
6	**(d)** 21 cm	$2k(4 + 7) = 66$ gives $k = 3$
7	**(a)** 60 cm	AO = OC; find x first, then AC = 2 AO
8	**(c)** 18	Adjacent angles are supplementary
9	**(b)** 102°	$\angle B = \angle D$ and the angle sum is 360°
10	**(c)** 75°	Exterior angles of a quadrilateral add up to 360°

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	AO = OC gives $x = 4$ (1); AO $= 22$ cm so AC $= 44$ cm (1).	1+1
12	Fourth exterior $= 360 - 278 = 82^\circ$ (1); interior $= 180 - 82 = 98^\circ$ (1).	1+1
13	One part $= 360 \div 24 = 15^\circ$ (1); largest $= 150^\circ$ (1).	1+1
14	AO = OC gives $x = 4$ (1); AO $= 29$ cm so AC $= 58$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4x + 72 = 360$ (1); $x = 72$ (1); angles 72°, 84°, 96°, 108° (1).	1+1+1
16	Sides $= 5k$ and $7k$; $2(5k + 7k) = 72$ (1); $k = 3$ (1); sides 15 cm and 21 cm (1).	1+1+1
17	$\angle CAB = 90 - 38 = 52^\circ$ (1); OA = OB so $\angle AOB = 180 - 2(52) = 76^\circ$ (1); OC = OD so $\angle ODC = \angle OCD = \angle CAB = 52^\circ$ (1).	1+1+1
	OR: $4x + 108 = 360$ (1); $x = 63$ (1); angles 63°, 81°, 99°, 117° (1).	1+1+1
18	Sides $= 4k$ and $9k$; $2(4k + 9k) = 52$ (1); $k = 2$ (1); sides 8 cm and 18 cm (1).	1+1+1
	OR: $\angle CAB = 90 - 52 = 38^\circ$ (1); OA = OB so $\angle AOB = 180 - 2(38) = 104^\circ$ (1); OC = OD so $\angle ODC = \angle OCD = \angle CAB = 38^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diagonals meet at right angles, so $\angle OAB + \angle OBA = 90^\circ$ (1); $4x + 2 + 5x - 2 = 90$ gives $x = 10$ (2); $\angle A = 2 \times 42 = 84^\circ$ (1); $\angle B = 2 \times 48 = 96^\circ$ (1).	1+2+1+1
	OR: Diagonals of a rectangle are equal: $4x + 6 = 2x + 16$ gives $x = 5$ (2); AC = BD $= 26$ cm (1); OA $= 13$ cm (1); wire $= 2 \times 26 = 52$ cm (1).	2+1+1+1
20	Diagonals meet at right angles, so $\angle OAB + \angle OBA = 90^\circ$ (1); $2x + 20 + 4x - 8 = 90$ gives $x = 13$ (2); $\angle A = 2 \times 46 = 92^\circ$ (1); $\angle B = 2 \times 44 = 88^\circ$ (1).	1+2+1+1
	OR: Diagonals of a rectangle are equal: $6x + 1 = 4x + 11$ gives $x = 5$ (2); AC = BD $= 31$ cm (1); OA $= 15.5$ cm (1); wire $= 2 \times 31 = 62$ cm (1).	2+1+1+1

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).

- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** are perpendicular	Perpendicular diagonals
2	**(c)** 81	$4x + 36 = 360$
3	**(c)** 30 cm	$2k(5 + 6) = 110$ gives $k = 5$
4	**(a)** 44 cm	AO = OC; find x first, then AC = 2 AO
5	**(a)** 16	Adjacent angles are supplementary
6	**(c)** 95°	$\angle B = \angle D$ and the angle sum is 360°
7	**(b)** 85°	Exterior angles of a quadrilateral add up to 360°
8	**(c)** Every rhombus is a parallelogram	A rhombus has both pairs of opposite sides parallel
9	**(a)** 69	$4x + 84 = 360$
10	**(c)** 36 cm	$2k(1 + 4) = 90$ gives $k = 9$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fourth exterior $= 360 - 253 = 107^\circ$ (1); interior $= 180 - 107 = 73^\circ$ (1).	1+1
12	One part $= 360 \div 20 = 18^\circ$ (1); largest $= 126^\circ$ (1).	1+1
13	AO = OC gives $x = 4$ (1); AO $= 22$ cm so AC $= 44$ cm (1).	1+1
14	Fourth exterior $= 360 - 284 = 76^\circ$ (1); interior $= 180 - 76 = 104^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4x + 144 = 360$ (1); $x = 54$ (1); angles 54°, 78°, 102°, 126° (1).	1+1+1
16	Sides $= 3k$ and $8k$; $2(3k + 8k) = 110$ (1); $k = 5$ (1); sides 15 cm and 40 cm (1).	1+1+1
17	$\angle CAB = 90 - 22 = 68^\circ$ (1); OA = OB so $\angle AOB = 180 - 2(68) = 44^\circ$ (1); OC = OD so $\angle ODC = \angle OCD = \angle CAB = 68^\circ$ (1).	1+1+1
	OR: $4x + 24 = 360$ (1); $x = 84$ (1); angles 84°, 88°, 92°, 96° (1).	1+1+1
18	Sides $= 6k$ and $7k$; $2(6k + 7k) = 104$ (1); $k = 4$ (1); sides 24 cm and 28 cm (1).	1+1+1
	OR: $\angle CAB = 90 - 28 = 62^\circ$ (1); OA = OB so $\angle AOB = 180 - 2(62) = 56^\circ$ (1); OC = OD so $\angle ODC = \angle OCD = \angle CAB = 62^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diagonals meet at right angles, so $\angle OAB + \angle OBA = 90^\circ$ (1); $3x + 6x = 90$ gives $x = 10$ (2); $\angle A = 2 \times 30 = 60^\circ$ (1); $\angle B = 2 \times 60 = 120^\circ$ (1).	1+2+1+1
	OR: Diagonals of a rectangle are equal: $7x + 4 = 5x + 16$ gives $x = 6$ (2); AC = BD $= 46$ cm (1); OA $= 23$ cm (1); wire $= 2 \times 46 = 92$ cm (1).	2+1+1+1
20	Diagonals meet at right angles, so $\angle OAB + \angle OBA = 90^\circ$ (1); $7x - 3 + 2x + 3 = 90$ gives $x = 10$ (2); $\angle A = 2 \times 67 = 134^\circ$ (1); $\angle B = 2 \times 23 = 46^\circ$ (1).	1+2+1+1
	OR: Diagonals of a rectangle are equal: $8x + 3 = 5x + 18$ gives $x = 5$ (2); AC = BD $= 43$ cm (1); OA $= 21.5$ cm (1); wire $= 2 \times 43 = 86$ cm (1).	2+1+1+1

Mistakes to avoid:

- Saying the diagonals of a rhombus are equal, or the diagonals of a rectangle are perpendicular; it is the other way round.
- Treating adjacent angles of a parallelogram as equal; **opposite** angles are equal and adjacent angles add up to 180°.
- Forgetting that every square is a rectangle and a rhombus, and every rectangle or rhombus is a parallelogram.
- Using the sum 180° (triangle) instead of 360° for the interior angles of a quadrilateral.
- Saying a trapezium has two pairs of parallel sides; only one pair is parallel (two pairs make a parallelogram).

- Assuming a quadrilateral with equal diagonals must be a rectangle; the diagonals must also bisect each other, and a parallelogram is needed first.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND